Day 101: Text Summarization — Extractive vs Abstractive, on Real News

MD Mutasim Billah | 52-Week Data Science to ML/AI Roadmap | Week 15, Day 4 | Complete study guide

Watch: Text Summarization Explained: Abstractive vs Extractive (SystemDR) | Abstractive Text Summarization using Transformers-BART Model (ProjectPro)

Script: `day101_summarization.py` · every number in this guide is copied by program from `day101_run_output.txt`, the real output of that script.

Four summaries of 100 real news articles, scored with Day 97's ROUGE against the journalists' highlights. Copying the first three sentences (lead-3, ROUGE-2 0.166) is not beaten by a pretrained T5 that writes its own summary (0.154; the difference is inside the noise) and clearly beats TextRank written from scratch (0.122). ROUGE also misses what matters most about T5: 15 of its 100 summaries contain a number or word the article does not support, and reading them finds real errors — 'Friday' for Thursday, 'the Italian capital' for Madrid, an invented author, and 'blackburn beat blackburn'.

0. Where today fits

DEFINITION — start here, no prior knowledge assumed

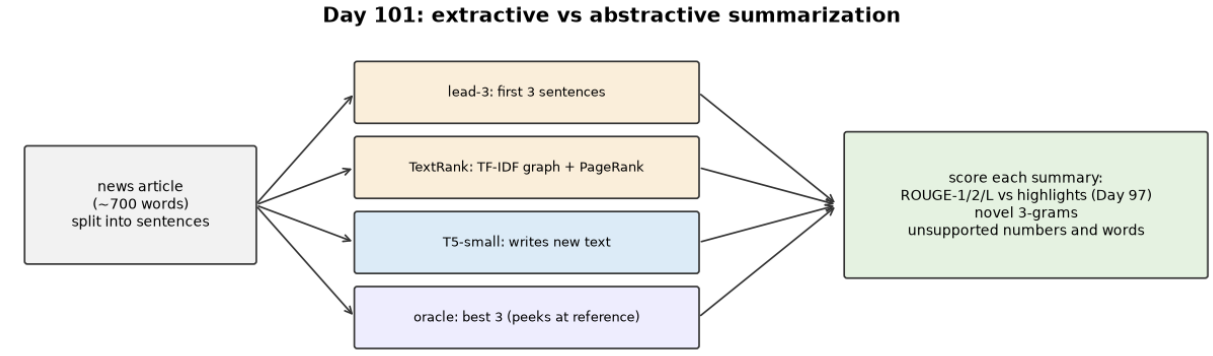
Extractive summarisation chooses whole sentences from the text and copies them unchanged. **Abstractive** summarisation writes a new, shorter text with a sequence-to-sequence model, which may use words and claims the source never contains.

T5 (text-to-text transfer Transformer) treats every task as text in, text out; prefixing an article with 'summarize: ' asks for a summary. T5-small has 60 million weights and reads at most 512 tokens.

REAL-WORLD SCENARIO

A highlighter (extractive: the words are guaranteed to be the author's) against a note-taker who rewrites in their own words (abstractive: shorter, smoother, and sometimes wrong).

SYSTEM DIAGRAM



Four ways to summarise one article, and the checks applied to each summary.

Today's part for Day 104

`day101_exercise.py` is the text-cutting part: `split_sentences` (rules for full stops, abbreviations and initials) and `chunk_words` (overlapping word windows, which Day 104 searches). Its real output:

ACTUAL OUTPUT

```
| Mr. Smith arrived in Washington on Monday.
  | He met Dr. J. R. Doe at 5 p.m.
  | "It went well," he said.
  | U.S. officials declined to comment.
  | Talks resume in Oct. 2025, a spokesman said.
5 sentences (expected 5)
size 10, overlap 3: 3 chunks of [10, 10, 9] words; all 23 words covered: True; words shared by
    neighbours: [3, 3]
size 8, overlap 0: 3 chunks of [8, 8, 7] words; all 23 words covered: True; words shared by
    neighbours: [0, 0]
size 30, overlap 5: 1 chunks of [23] words; all 23 words covered: True; words shared by
    neighbours: []
```

How to read each section

Step	What it gives you
DEFINITION	What the idea is, in plain words. Start here.
REAL-WORLD SCENARIO	An everyday situation where the same idea shows up.
MATHEMATICS	The formula, when the section introduces one.
IMPLEMENTATION	The real code from the script.
ACTUAL OUTPUT	What the script really printed, copied unchanged.
What it means	How to read the numbers, and what they do and do not show.
Why choose this? / Why NOT this?	When the idea is the right tool, and what it costs.

1. The data: real news and real highlights

DEFINITION — start here, no prior knowledge assumed

100 articles drawn at random from the CNN/DailyMail test set, each with the bullet-point highlights its journalists wrote, used as the reference summary. Sentences are found by rule (Day 101's exercise part).

REAL-WORLD SCENARIO

Comparing a student's summary of a news story with the summary box the newspaper printed.

ACTUAL OUTPUT

```
articles: 699 words on average (10th-90th percentile 272-1191), 31.8 sentences; reference highlights: 52 words
first article's opening sentences (rule-based splitting):
  | Boy, would Tiger like to find a fairway here,’ said former professional David Feherty.
  | ‘Preferably one attached to this hole.’ It was that sort of day.
  | Tiger Woods had worked so hard on his short game, the other quality needed to take a safe passage through
      these parts ha
its reference: The Masters 2015 crowd wanted Tiger Woods to do well at Augusta. He was still the 111th ranked
    golfer in the world. A bit erratic, a bit brilliant, a bit rusty, a bit fallible. And, fleetingly, a bit
    marvellous too. The gallery was 10, 12 d
Actual output from this lesson's run.
```

What it means

Articles average 699 words (31.8 sentences); references average 52. The rule-based splitter handles the curly quotes; the first article starts mid-quote, exactly as stored in the dataset.

WHY CHOOSE THIS?

A real benchmark with human-written references.

WHY NOT THIS? (TRADEOFFS)

One reference per article; many good summaries would differ from it.

2. Four summaries of each article

DEFINITION — start here, no prior knowledge assumed

Lead-3: the first three sentences. **TextRank**: sentences become TF-IDF vectors, each sentence 'votes' for similar ones, PageRank finds the most central three. **T5-small**: generated with the settings saved in its configuration (beam search, no repeated 3-grams). **Oracle**: the three sentences that score best against the reference — it looks at the answer, so it is an upper bound, not a system.

REAL-WORLD SCENARIO

Four ways to tell a friend about an article: read the opening, pick the most 'central' sentences, retell it, or copy the ones you know the teacher wants.

MATHEMATICS

$$\text{TextRank: } r \leftarrow (1-d)/n + d\,W^{T} r, \quad W_{ij} = \cos(s_i, s_j) \,/\, \Sigma_j \cos(s_i, s_j), \quad d = 0.85$$

IMPLEMENTATION

```
def textrank(sents, k=K, d=0.85, iters=100):
    V = tfidf(sents); S = V @ V.T
    np.fill_diagonal(S, 0)
    W = S / (S.sum(1, keepdims=True) + 1e-12)
    r = np.ones(len(sents)) / len(sents)
    for _ in range(iters):
        r = (1 - d) / len(sents) + d * W.T @ r
    top = sorted(np.argsort(-r)[:k])
    return " ".join(sents[i] for i in top), r
```

ACTUAL OUTPUT

```
T5-small: 60,506,624 weights; settings from its config: {'early_stopping': True, 'length_penalty': 2.0, 'max_length': 200,
    'min_length': 30, 'no_repeat_ngram_size': 3, 'num_beams': 4, 'prefix': 'summarize: '}
  7.2 s per article on this CPU (4 threads); input cut at 512 tokens
the first article, each system:
  lead-3   | Boy, would Tiger like to find a fairway here,' said former professional David Feherty. 'Preferably one attached
      to this hole.' It was that sort of day. Tiger Woods had worked so hard on his short game, the other quality needed to
  TextRank | He walked to the ball, perched at the top of a small incline to the right of the green. There was a shot on the
      seventh that would have been a contender for play of the day. It was the golf of a competitor still battling against t
  T5-small | the world no 111 struggled to get under par on his first round at augusta. woods had a poor chip shot on the
      third hole. the former professional said he would like to find a fairway here. he said: 'i think i'm not a golfer, but
      i
  oracle   | Still the biggest, still primed to applaud even the tiniest glimmer of his genius. But he was still the 111th
      ranked golfer in the world. A bit rusty, a bit erratic, a bit brilliant, a bit fallible.
Actual output from this lesson's run.
```

What it means

Lead-3 and the oracle are real sentences; T5 rewrites. On the first article T5 already writes a quotation ('i think i'm not a golfer, but i don't know what to do') using words the article never contains (section 4). T5-small took 7.2 s per article on this machine, with other jobs running.

WHY CHOOSE THIS?

Extractive summaries are fast and can never misquote.

WHY NOT THIS? (TRADEOFFS)

They can be choppy, and long when sentences are long.

3. ROUGE against the journalists' highlights

DEFINITION — start here, no prior knowledge assumed

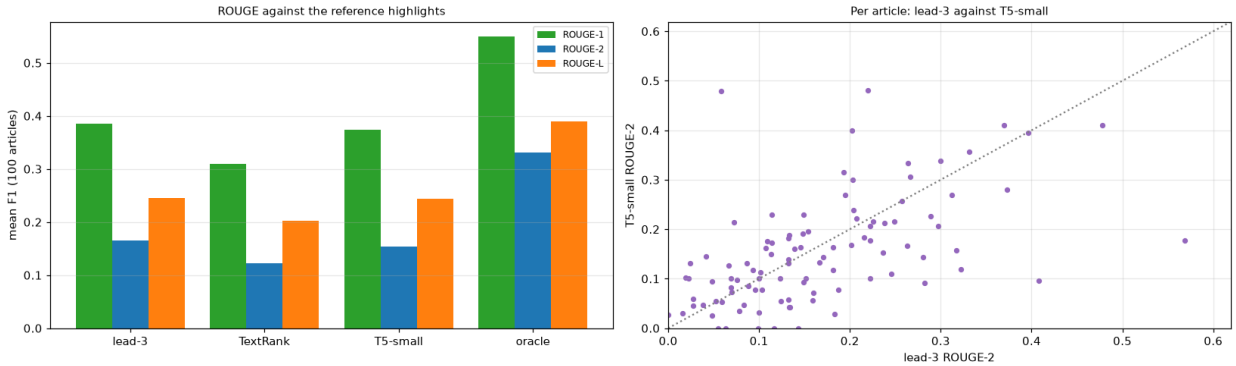
Day 97's ROUGE-1, ROUGE-2 and ROUGE-L F1 for every article, averaged, with Day 99's bootstrap interval for ROUGE-2 and paired comparisons on the same articles.

REAL-WORLD SCENARIO

Counting how many of the newspaper's own phrases each summary uses.

ACTUAL OUTPUT

```
Mean F1 over 100 articles (Day 97's functions; words are lowercase letters only, so numbers are ignored), with Day 99's 95%
    bootstrap interval for ROUGE-2, and the mean length.
system      ROUGE-1  ROUGE-2  ROUGE-L    ROUGE-2 95% interval  words
lead-3        0.386    0.166    0.246      [0.144, 0.187]     78
TextRank      0.310    0.122    0.203      [0.102, 0.145]    109
T5-small      0.374    0.154    0.244      [0.134, 0.175]     46
oracle        0.550    0.331    0.390      [0.301, 0.366]     61
paired differences in ROUGE-2 (Day 99's paired bootstrap): mean, 95% interval, share of resamples ahead
  lead-3    minus TextRank  +0.043 [+0.018, +0.067]  ahead in 1.000
  lead-3    minus T5-small  +0.011 [-0.009, +0.032]  ahead in 0.876
  T5-small  minus TextRank  +0.032 [+0.004, +0.056]  ahead in 0.988
  oracle    minus lead-3    +0.166 [+0.140, +0.195]  ahead in 1.000
Actual output from this lesson's run.
```



Left: mean ROUGE by system. Right: ROUGE-2 per article, lead-3 against T5-small.

What it means

Lead-3 is hard to beat on news, where journalists put the key facts first: ROUGE-2 0.166. T5-small (0.154) is level with it (difference +0.011, interval crossing 0) with summaries well under two-thirds as long (46 words against 78). TextRank, from scratch, is clearly worst (0.122): the most 'central' sentences of an article are not its news. The oracle's 0.331 shows how much better sentence choice could be. The scatter shows the per-article picture: neither system wins consistently.

WHY CHOOSE THIS?

Paired comparisons on the same articles make small differences testable.

WHY NOT THIS? (TRADEOFFS)

Numbers are ignored by Day 97's word tokenizer, and ROUGE rewards overlap, not truth.

4. What ROUGE cannot see: new wording and unsupported details

DEFINITION — start here, no prior knowledge assumed

Novel 3-grams: the share of a summary's word triples that never occur in the article. **Unsupported details**: numbers, and words of four or more letters, in the summary with no word in the article sharing their first five letters (so 'suffered' is supported by 'suffer'). A flag is a prompt to read, not a verdict, so the script prints the flagged sentence next to the article's opening.

REAL-WORLD SCENARIO

Fact-checking a retold story: anything said that was not in the original needs a source.

IMPLEMENTATION

```
def novel_trigrams(summary, article):
    s, a = d97.words(summary), set(d97.ngrams(d97.words(article), 3))
    tri = [tuple(s[i:i + 3]) for i in range(len(s) - 2)]
    return np.mean([t not in a for t in tri]) if tri else 0.0
def unsupported(summary, article):
    nums = set(re.findall(r"\d[\d,.]*\d|\d", summary)) - set(re.findall(r"\d[\d,.]*\d|\d",
        article))
    stems = {w[:5] for w in re.findall(r"[a-z]+", article.lower())}
    new = {w for w in re.findall(r"[a-z]+", summary.lower()) if len(w) >= 4 and w[:5] not in
        stems}
    return nums | new
```

ACTUAL OUTPUT

```
Novel 3-grams: share of the summary's word triples that never occur in the article (0 = pure copying). Unsupported: numbers,
    and words of four or more letters, in the summary whose first five letters match no word of the article (lowercase,
    letters only, since T5-small often lowercases words; 'suffered' counts as supported by 'suffer').
system      novel 3-grams    summaries with an unsupported detail
lead-3              0.000                                    0
TextRank            0.033                                    0
T5-small            0.235                                   15
T5-small's unsupported details, all 15 cases: article 0: ['know', 'think']; article 7: ['23']; article 14: ['italian'];
    article 16: ['friday']; article 19: ['beat']; article 21: ['match']; article 27: ['year']; article 30: ['comes']; article
    34: ['latest', 'manchester', 'news', 'united']; article 57: ['posted']; article 63: ['avlon', 'john']; article 65:
    ['pick']; article 70: ['saturday', 'says']; article 82: ['killed']; article 95: ['girl']
each case: the T5 sentence holding the first flagged item, and the article's opening
  article  0 [know] T5: he said: 'i think i'm not a golfer, but i don't know what to do'
             article: Boy, would Tiger like to find a fairway here,' said former professional David Feherty. 'Preferably one
                 attached to this hole.' It was that sort of day
  article  7 [23] T5: the 23-year-old will be with them throughout their Blast campaign.
             article: Surrey have signed Australia all-rounder Moises Henriques for this summer's NatWest T20 Blast.
                 Henriques will join Surrey immediately after the Indian
  article 14 [italian] T5: but he admits he has heard about interest from the italian capital.
             article: Arsenal midfielder Santi Cazorla has said he is happy to stay at Arsenal despite interest from Atletico
                 Madrid, although he admits that 'you never kno
  article 16 [friday] T5: a 31-year-old woman was arrested in queens, new york on friday.
             article: Asia Siddiqui not only saw herself as a soldier for Allah, but a poet as well. The 31-year-old woman
                 who was arrested Thursday in New York for alleged
  article 19 [beat] T5: blackburn beat blackburn 1-0 in their quarter-final replay at Ewood Park.
             article: It is not often you hear this said but the goalkeeper really should have scored. Deep in injury time,
                 with Blackburn's FA Cup journey about to end, Si
  article 21 [match] T5: he was carried off on a stretcher during the match at the amex stadium.
             article: Liverpool starlet Joao Teixeira has been ruled out for the rest of the season after suffering a broken
                 leg during his loan spell at Brighton. The 22-y
  article 27 [year] T5: Every house in the village got one as far as we are aware,' said 67-year-old.
             article: A mysterious Easter Bunny has delivered chocolate eggs to every house in a village of 3,000 people.
                 Residents from Alderholt, Dorset, woke to discover
  article 30 [comes] T5: it comes as evidence against slager continues to mount.
             article: Killer cop Michael Slager is not eligible for South Carolina's death penalty even if he is convicted of
```

```
                murder, according to local prosecutors. The No
  article 34 [latest] T5: click here for all the latest manchester united news.
             article: Andros Townsend hailed Harry Kane as the 'best finisher' he has played with and predicted a bright
                 international career for his Tottenham Hotspur team
  article 57 [posted] T5: he posted an image of himself alongside four of his former colleagues and their respective partners
      at a restaurant table.
             article: Ryan Giggs, Paul Scholes, Gary Neville, Nicky Butt and Andrew Cole went out for dinner just hours after
                 their former side's 1-0 defeat by Chelsea. Ex-
  article 63 [avlon] T5: john avlon: bernie Sanders is a long shot to win election to the highest office in the nation.
             article: (CNN)Bernie Sanders, my Vermont senator and, indeed, a friend of many years, is now running for
                 president.  He noted at his announcement (with a famil
  article 65 [pick] T5: the referee for Floyd Mayweather's clash with Manny Pacquiao will earn $10,000 (£6,800) the fight at
      the MGM Grand on may 2 is expected to draw revenu
             article: The referee for Floyd Mayweather's clash with Manny Pacquiao in Las Vegas next month will earn $10,000
                 (£6,800). The fight at the MGM Grand on May 2 i
  article 70 [saturday] T5: "Today I'm at a loss for words but I just remember Bob being a funny guy," the band's official
      Facebook page says saturday.
             article: Atlanta (CNN)Robert Lewis Burns Jr., the original drummer in Southern rock band Lynyrd Skynyrd, died
                 Friday night in a car crash, a Georgia State Patr
  article 82 [killed] T5: now 41, Landon has visited the crash scene and is thought to be the first child survivor to return
      to the countryside where the crash killed 78 childr
             article: An orphan who survived a horrific plane crash that claimed the lives of nearly 80 babies and children
                 40 years ago  has made an emotional return to th
  article 95 [girl] T5: 'debate me like one of your French girls' - a reference to when Rose asks Jack in the 1997 film
      Titanic to 'draw me like a french girl'.
             article: From David Cameron and Nick Clegg sitting on the Gogglebox sofa to Ed Miliband doing a Kate Winslet in
                 Titanic, these are some of the best memes poste
Actual output from this lesson's run.
```

What it means

Extractive summaries copy (0.000 and 0.033 novel 3-grams) and have nothing to flag. T5-small writes 0.235 new 3-grams, and 15 of its summaries are flagged. Reading the printed evidence sorts them. Several are harmless rewording ('comes', 'posted', 'year'). Several are real errors: article 16's arrest happened on Thursday, T5 says Friday; article 14's interest came from Atletico Madrid, T5 says 'the italian capital'; article 63, an opinion piece by a friend of Bernie Sanders, gets an invented byline, 'john avlon:'; article 34 ends with 'click here for all the latest manchester united news', boilerplate from the kind of page T5 was trained on; article 19 says 'blackburn beat blackburn'; article 0 quotes words ('know', 'think') the article never uses. The check also misses things: an error built from words the article does contain (a wrong number of victims, a swapped name) passes.

WHY CHOOSE THIS?

A cheap filter that points a human at the riskiest sentences.

WHY NOT THIS? (TRADEOFFS)

Neither complete nor exact: it flags paraphrase and misses errors made of the article's own words.

Interview preparation — technical questions

Q: What is the lead-3 baseline?

A: The first three sentences; on news it scored ROUGE-2 0.166, level with T5-small.

Q: How does TextRank choose sentences?

A: PageRank on a sentence-similarity graph; the most central sentences win.

Q: Why is the oracle not a real system?

A: It chooses sentences by looking at the reference: an upper bound (0.331).

Q: How do you compare two summarisers fairly?

A: Per-document scores and a paired bootstrap on the same documents.

Q: How can you detect hallucinated details automatically?

A: Check numbers and words against the source; then read the flagged sentences.

Interview preparation — theoretical questions

Q: Why is lead-3 strong on news?

A: News is written with the key facts first (the inverted pyramid).

Q: Why can abstractive models invent facts?

A: They generate likely text; nothing forces every claim to come from the input.

Q: Why doesn't ROUGE catch it?

A: It rewards word overlap; a wrong day or a wrong city costs one word.

Q: When would you choose extractive?

A: When every sentence must be traceable to the source (legal, medical, news).

Key takeaway and what's next

> **KEY TAKEAWAY**
>
> On real news, copying the opening is as good as a pretrained abstractive model by ROUGE (0.166 against 0.154), and both beat a from-scratch TextRank (0.122). Abstractive summaries are shorter and read naturally, but 15 of 100 were flagged, and reading them found a wrong day, a wrong city, an invented author and nonsense. Score summaries with ROUGE, then read them.

- ROUGE-1/2/L: lead-3 0.386/0.166/0.246; TextRank 0.310/0.122/0.203; T5-small 0.374/0.154/0.244; oracle 0.550/0.331/0.390.
- Lead-3 minus T5-small, ROUGE-2: +0.011 [-0.009, +0.032].
- Novel 3-grams: T5-small 0.235; summaries with an unsupported detail: 15 of 100.

Suggested watch order, if starting from zero

- 1. Text Summarization Explained: Abstractive vs Extractive (SystemDR) — the two kinds of summarisation.
- 2. Abstractive Text Summarization using Transformers-BART Model (ProjectPro) — abstractive summarisation with a Transformer in practice.

Day 102 preview

- **Question answering.** A pretrained model that points at the answer inside a Wikipedia paragraph, scored on real SQuAD questions with the official metrics written from scratch.